

ECON3350/7350 Applied Econometrics for Macroeconomics and Finance

Revision Lecture

Fu Ouyang

University of Queensland

Reference Materials

Author	Title	Call No
	Lecture Notes Combined with your own notes from listening and readings	
	Tutorials and Assignments	
Enders, W	Applied Econometric Time Series, 4e	HB139 .E55 2015
Verbeek, M	A Guide to Modern Econometrics	HB139 .V465 2012

Topics Covered

1. Revision of Regression Analysis

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II
3. Single Equation Models of Multiple Time Series - ARDL

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II
3. Single Equation Models of Multiple Time Series - ARDL
4. Deterministic and Stochastic Trends

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II
3. Single Equation Models of Multiple Time Series - ARDL
4. Deterministic and Stochastic Trends
5. Cointegration - The Single Equation Model

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II
3. Single Equation Models of Multiple Time Series - ARDL
4. Deterministic and Stochastic Trends
5. Cointegration - The Single Equation Model
6. Volatility Models

Topics Covered

1. Revision of Regression Analysis
2. Univariate Time Series - I and II
3. Single Equation Models of Multiple Time Series - ARDL
4. Deterministic and Stochastic Trends
5. Cointegration - The Single Equation Model
6. Volatility Models
7. SVAR, VAR, VEC Models: - The Multiple Equation Model

Regression Analysis Review

► Linear Model

Regression Analysis Review

- ▶ Linear Model
- ▶ Estimation - Least Squares

Regression Analysis Review

- ▶ Linear Model
- ▶ Estimation - Least Squares
- ▶ Goodness of Fit

Regression Analysis Review

- ▶ Linear Model
- ▶ Estimation - Least Squares
- ▶ Goodness of Fit
- ▶ Inference - t and F tests

Regression Analysis Review

- ▶ Linear Model
- ▶ Estimation - Least Squares
- ▶ Goodness of Fit
- ▶ Inference - t and F tests
- ▶ Heteroskedasticity and Autocorrelation

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**
 - ▶ **Model Selection:** SACF, SPACF to identify the models, AIC and BIC

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**
 - ▶ **Model Selection:** SACF, SPACF to identify the models, AIC and BIC
 - ▶ **Estimation, Diagnostics:** residual time series plot, ACF, Q-statistic (Ljung-Box test) to test for residual autocorrelation.

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**
 - ▶ **Model Selection:** SACF, SPACF to identify the models, AIC and BIC
 - ▶ **Estimation, Diagnostics:** residual time series plot, ACF, Q-statistic (Ljung-Box test) to test for residual autocorrelation.
- ▶ **Forecasting:** Conditional expectation of future values given the observed sample.

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**
 - ▶ **Model Selection:** SACF, SPACF to identify the models, AIC and BIC
 - ▶ **Estimation, Diagnostics:** residual time series plot, ACF, Q-statistic (Ljung-Box test) to test for residual autocorrelation.
- ▶ **Forecasting:** Conditional expectation of future values given the observed sample.
 - ▶ One-step ahead, h-steps ahead predictions and standard errors.

Univariate Time Series Models

- ▶ **Basic Concepts:** Time series, stochastic process, realization, moments, stationarity
- ▶ **Common Time Series Models:** $AR(p)$, $MA(q)$, $ARMA(p, q)$
- ▶ **Theoretical ACF, PACF:** Autocovariance, Autocorrelation, Autocorrelation Function, Partial Autocorrelation Function.
- ▶ **Fitting Univariate ARMA(p,q) models:**
 - ▶ **Model Selection:** SACF, SPACF to identify the models, AIC and BIC
 - ▶ **Estimation, Diagnostics:** residual time series plot, ACF, Q-statistic (Ljung-Box test) to test for residual autocorrelation.
- ▶ **Forecasting:** Conditional expectation of future values given the observed sample.
 - ▶ One-step ahead, h-steps ahead predictions and standard errors.
- ▶ **MA Representation - Impulse Responses:** Write ARMA(p,q) models as an $MA(\infty)$ and then solve for the impulse response function.

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models
- ▶ Dynamic Responses. The ARDL Multipliers: Impact Multiplier, Intermediate Multipliers, Long-run multiplier

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models
- ▶ Dynamic Responses. The ARDL Multipliers: Impact Multiplier, Intermediate Multipliers, Long-run multiplier
- ▶ The ARDL Family of Models

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models
- ▶ Dynamic Responses. The ARDL Multipliers: Impact Multiplier, Intermediate Multipliers, Long-run multiplier
- ▶ The ARDL Family of Models
 - ▶ Error Correction Mechanism and Error Correction Model

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models
- ▶ Dynamic Responses. The ARDL Multipliers: Impact Multiplier, Intermediate Multipliers, Long-run multiplier
- ▶ The ARDL Family of Models
 - ▶ Error Correction Mechanism and Error Correction Model
 - ▶ Testing Restrictions on the ADL: $f = \frac{(S_0 - S_1)/J}{S_1/(T - K)} \sim F_{J, (T-K)}$

Single Equation Models

- ▶ Common Time Series Models: $AR(p)$, $ARDL$, $MA(q)$, $ARMA(p, q)$
- ▶ The Autoregressive Distributed Lag Model: A general class of models
- ▶ Dynamic Responses. The ARDL Multipliers: Impact Multiplier, Intermediate Multipliers, Long-run multiplier
- ▶ The ARDL Family of Models
 - ▶ Error Correction Mechanism and Error Correction Model
 - ▶ Testing Restrictions on the ADL: $f = \frac{(S_0 - S_1)/J}{S_1/(T - K)} \sim F_{J, (T-K)}$
 - ▶ Use AIC, BIC and residual diagnostics as in regression analysis.

Trends

- ▶ **Non-stationarity:** Mean, Variance, Covariance

Trends

- ▶ **Non-stationarity**: Mean, Variance, Covariance
 - ▶ Trend Stationary vs Difference Stationary

Trends

- ▶ **Non-stationarity:** Mean, Variance, Covariance
 - ▶ Trend Stationary vs Difference Stationary
 - ▶ Stochastic Trends, Unit Roots and Common trends

Trends

- ▶ **Non-stationarity:** Mean, Variance, Covariance
 - ▶ Trend Stationary vs Difference Stationary
 - ▶ Stochastic Trends, Unit Roots and Common trends
 - ▶ Integrated Processes

Trends

- ▶ **Non-stationarity:** Mean, Variance, Covariance
 - ▶ Trend Stationary vs Difference Stationary
 - ▶ Stochastic Trends, Unit Roots and Common trends
 - ▶ Integrated Processes
 - ▶ ARIMA Models

Trends

- ▶ **Non-stationarity:** Mean, Variance, Covariance
 - ▶ Trend Stationary vs Difference Stationary
 - ▶ Stochastic Trends, Unit Roots and Common trends
 - ▶ Integrated Processes
 - ▶ ARIMA Models
- ▶ **Testing for a Unit Root:** Dickey-Fuller/Augmented Dickey Fuller (Testing Strategy)

Cointegration

- ▶ Spurious Regressions vs Cointegrated Regressions

Cointegration

- ▶ Spurious Regressions vs Cointegrated Regressions
- ▶ Cointegration:

Cointegration

- ▶ Spurious Regressions vs Cointegrated Regressions
- ▶ Cointegration:
 - ▶ Definition

Cointegration

- ▶ Spurious Regressions vs Cointegrated Regressions
- ▶ Cointegration:
 - ▶ Definition
 - ▶ Common Stochastic Trends

Cointegration

- ▶ Spurious Regressions vs Cointegrated Regressions
- ▶ Cointegration:
 - ▶ Definition
 - ▶ Common Stochastic Trends
 - ▶ Regression based approach to testing for cointegration (Engle-Granger)

Volatility Models

► Volatility and Non-Linear Models

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance
 - ▶ **Models:** Linear and ARIMA Models with ARCH and GARCH errors

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance
 - ▶ **Models:** Linear and ARIMA Models with ARCH and GARCH errors
 - ▶ **Fitting Models:** AIC, BIC, Residuals and Squared Residuals, LM Tests for ARCH effects

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance
 - ▶ **Models:** Linear and ARIMA Models with ARCH and GARCH errors
 - ▶ **Fitting Models:** AIC, BIC, Residuals and Squared Residuals, LM Tests for ARCH effects
 - ▶ Forecasting Volatility

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance
 - ▶ **Models:** Linear and ARIMA Models with ARCH and GARCH errors
 - ▶ **Fitting Models:** AIC, BIC, Residuals and Squared Residuals, LM Tests for ARCH effects
 - ▶ Forecasting Volatility
 - ▶ **Extensions:** EGARCH, TGARCH, ARCH-in Mean

Volatility Models

- ▶ Volatility and Non-Linear Models
- ▶ Conditional Heteroskedasticity
 - ▶ ARCH and GARCH models of the variance
 - ▶ **Models:** Linear and ARIMA Models with ARCH and GARCH errors
 - ▶ **Fitting Models:** AIC, BIC, Residuals and Squared Residuals, LM Tests for ARCH effects
 - ▶ Forecasting Volatility
 - ▶ **Extensions:** EGARCH, TGARCH, ARCH-in Mean
- ▶ Stochastic Volatility Models

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form
- ▶ **Vector Moving Average Representation**

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form
- ▶ **Vector Moving Average Representation**
 - ▶ Impulse Response Function and Forecast Error Variance Decomposition: Cholesky and Structural Decomposition

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form
- ▶ **Vector Moving Average Representation**
 - ▶ Impulse Response Function and Forecast Error Variance Decomposition: Cholesky and Structural Decomposition
 - ▶ Stability of VARs - Roots of Characteristic Polynomial, eigenvalues

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form
- ▶ **Vector Moving Average Representation**
 - ▶ Impulse Response Function and Forecast Error Variance Decomposition: Cholesky and Structural Decomposition
 - ▶ Stability of VARs - Roots of Characteristic Polynomial, eigenvalues
- ▶ **Lag Length** - Empirical Determination, AIC, BIC, LR, No Residual Autocorrelation

VAR Models

- ▶ **Multivariate - Multiple Equation Models.** All Variables are endogenous
- ▶ **Vector Autoregressive Model:** Structural and Reduced form
 - ▶ Dimension, Lags
 - ▶ Companion form
- ▶ **Vector Moving Average Representation**
 - ▶ Impulse Response Function and Forecast Error Variance Decomposition: Cholesky and Structural Decomposition
 - ▶ Stability of VARs - Roots of Characteristic Polynomial, eigenvalues
- ▶ **Lag Length** - Empirical Determination, AIC, BIC, LR, No Residual Autocorrelation
- ▶ **Granger Non-Causality.**

Cointegration - VECs

- ▶ Cointegration in a multiple equation setting

Cointegration - VECs

- ▶ Cointegration in a multiple equation setting
- ▶ Vector Error Correction Model: VAR(p), VEC.

Cointegration - VECs

- ▶ Cointegration in a multiple equation setting
- ▶ Vector Error Correction Model: VAR(p), VEC.
 - ▶ Cointegration Vectors, Cointegration Rank, Speed of Adjustment

Cointegration - VECs

- ▶ **Cointegration** in a multiple equation setting
- ▶ **Vector Error Correction Model**: VAR(p), VEC.
 - ▶ Cointegration Vectors, Cointegration Rank, Speed of Adjustment
- ▶ **Testing**: Engle-Granger vs Johansen Approach to Testing for the Cointegration rank (Trace and MaxEig statistics)

Cointegration - VECs

- ▶ **Cointegration** in a multiple equation setting
- ▶ **Vector Error Correction Model**: VAR(p), VEC.
 - ▶ Cointegration Vectors, Cointegration Rank, Speed of Adjustment
- ▶ **Testing**: Engle-Granger vs Johansen Approach to Testing for the Cointegration rank (Trace and MaxEig statistics)
- ▶ **Impulse Responses** and Permanent Shocks.

► Course and teaching evaluation

- ▶ Course and teaching evaluation
- ▶ For more information please see: <https://www.youtube.com/embed/videoseries?list=PLRvw9bTMoXagQGCxEi6hixS6szXYNIGBW>

- ▶ Course and teaching evaluation
- ▶ For more information please see: <https://www.youtube.com/embed/videoseries?list=PLRvw9bTMoXagQGCxEi6hixS6szXYNIGBW>
- ▶ You can access the surveys via the direct link in SECaTs email or via the survey portal at <https://eval.uq.edu.au>.

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).
4. Structure

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).
4. Structure
 - ▶ 10 MCQ + 3 problem solving questions.

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).
4. Structure
 - ▶ 10 MCQ + 3 problem solving questions.
 - ▶ Questions have many subparts.

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).
4. Structure
 - ▶ 10 MCQ + 3 problem solving questions.
 - ▶ Questions have many subparts.
 - ▶ Some questions include output with actual data

Examination Info

1. Time Allowed: TWO (2) hours (plus 10 minutes reading time)
2. Writing on rough paper (provided) during reading time.
3. Exam covers all topics in the course (Lecture 1 not directly).
4. Structure
 - ▶ 10 MCQ + 3 problem solving questions.
 - ▶ Questions have many subparts.
 - ▶ Some questions include output with actual data
 - ▶ Some subparts of the questions test concepts or ask for a derivation similar to assignments/tutorials.

Exam Preparation

- ▶ Revise every topic.

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.
 - ▶ Read and understand the concepts in lectures/tutorials. Make your own notes for each topic by combining all sources

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.
 - ▶ Read and understand the concepts in lectures/tutorials. Make your own notes for each topic by combining all sources
 - ▶ Assignment questions summarize key knowledge points, re-do them and make sure you 100% understand the solution

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.
 - ▶ Read and understand the concepts in lectures/tutorials. Make your own notes for each topic by combining all sources
 - ▶ Assignment questions summarize key knowledge points, re-do them and make sure you 100% understand the solution
 - ▶ Sample Example paper is available through Blackboard

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.
 - ▶ Read and understand the concepts in lectures/tutorials. Make your own notes for each topic by combining all sources
 - ▶ Assignment questions summarize key knowledge points, re-do them and make sure you 100% understand the solution
 - ▶ Sample Example paper is available through Blackboard
 - ▶ Attend consultation to check your answers with us.

Exam Preparation

- ▶ Revise every topic.
 - ▶ Lecture, tutorials and assignments.
 - ▶ Read and understand the concepts in lectures/tutorials. Make your own notes for each topic by combining all sources
 - ▶ Assignment questions summarize key knowledge points, re-do them and make sure you 100% understand the solution
 - ▶ Sample Example paper is available through Blackboard
 - ▶ Attend consultation to check your answers with us.
 - ▶ For the tests we have used commonly during the semester you must be able to state the null hypothesis, how the statistic is computed, which critical values you must use, how to state your decision.

Examples of What to Expect

- ▶ You are provided with plots of data, SACF, SPACF and unit root test output, and asked to identify possible univariate models (ARIMA(p,d,q)).

Examples of What to Expect

- ▶ You are provided with plots of data, SACF, SPACF and unit root test output, and asked to identify possible univariate models (ARIMA(p,d,q)).
- ▶ You are provided with output of several diagnostics and tests on residuals and squared residuals and asked to interpret and reach a conclusion on the model most suitable to continue the analysis.

Examples of What to Expect

- ▶ You are provided with plots of data, SACF, SPACF and unit root test output, and asked to identify possible univariate models (ARIMA(p,d,q)).
- ▶ You are provided with output of several diagnostics and tests on residuals and squared residuals and asked to interpret and reach a conclusion on the model most suitable to continue the analysis.
 - ▶ With the model chosen you are asked to construct a forecast

Examples of What to Expect

- ▶ You are provided with plots of data, SACF, SPACF and unit root test output, and asked to identify possible univariate models (ARIMA(p,d,q)).
- ▶ You are provided with output of several diagnostics and tests on residuals and squared residuals and asked to interpret and reach a conclusion on the model most suitable to continue the analysis.
 - ▶ With the model chosen you are asked to construct a forecast
- ▶ You are provided with Stata output of the estimation of a VAR or a VEC, impulse responses and/or FEVD, and asked to interpret and provide a conclusion

Examples of What to Expect

- ▶ You are provided with plots of data, SACF, SPACF and unit root test output, and asked to identify possible univariate models (ARIMA(p,d,q)).
- ▶ You are provided with output of several diagnostics and tests on residuals and squared residuals and asked to interpret and reach a conclusion on the model most suitable to continue the analysis.
 - ▶ With the model chosen you are asked to construct a forecast
- ▶ You are provided with Stata output of the estimation of a VAR or a VEC, impulse responses and/or FEVD, and asked to interpret and provide a conclusion
- ▶ You are provided with Stata output of the estimation of a VAR and asked to test for Granger Non-causality

Examples of What to Expect

- ▶ You are given an ARDL model and ask to write the long-run multiplier

Examples of What to Expect

- ▶ You are given an ARDL model and ask to write the long-run multiplier
- ▶ You are given an AR model and asked to write as an Augmented Dickey Fuller equation

Examples of What to Expect

- ▶ You are given an ARDL model and ask to write the long-run multiplier
- ▶ You are given an AR model and asked to write as an Augmented Dickey Fuller equation
- ▶ You are given a structural model and asked to derive the reduced form (companion form)

Examples of What to Expect

- ▶ You are given an ARDL model and ask to write the long-run multiplier
- ▶ You are given an AR model and asked to write as an Augmented Dickey Fuller equation
- ▶ You are given a structural model and asked to derive the reduced form (companion form)
- ▶ You are provided with Stata output of the estimation of several volatility models and asked to write the model that has been estimated and interpret the results.

Examples of What to Expect

- ▶ You are given an ARDL model and ask to write the long-run multiplier
- ▶ You are given an AR model and asked to write as an Augmented Dickey Fuller equation
- ▶ You are given a structural model and asked to derive the reduced form (companion form)
- ▶ You are provided with Stata output of the estimation of several volatility models and asked to write the model that has been estimated and interpret the results.
- ▶ You are provided with Stata output of testing for unit roots and asked to conduct the tests and conclude whether the series is integrated and what is the order of integration.

Examples of What to Expect

- ▶ You are provided with Stata output of testing for cointegration using the residuals of a regression and asked to conduct the test and conclude

Examples of What to Expect

- ▶ You are provided with Stata output of testing for cointegration using the residuals of a regression and asked to conduct the test and conclude
- ▶ You are provided with Stata output of testing for cointegration using the Johansen tests and asked to conduct the tests and conclude.

Examples of What to Expect

- ▶ You are provided with Stata output of testing for cointegration using the residuals of a regression and asked to conduct the test and conclude
- ▶ You are provided with Stata output of testing for cointegration using the Johansen tests and asked to conduct the tests and conclude.
- ▶ You are provided with estimated impulse responses obtained from a structural decomposition and a Cholesky decomposition and asked to explain why they differ.

Examples of What to Expect

- ▶ You are provided with Stata output of testing for cointegration using the residuals of a regression and asked to conduct the test and conclude
- ▶ You are provided with Stata output of testing for cointegration using the Johansen tests and asked to conduct the tests and conclude.
- ▶ You are provided with estimated impulse responses obtained from a structural decomposition and a Cholesky decomposition and asked to explain why they differ.
- ▶ You are provided with a VAR model and asked to explain how to test for Granger Non-Causality

Consultation

- ▶ The consultation timetable changes from next week. Please check BB announcement about Consultation Timetable

- ▶ All the best with your preparation and exams.

- ▶ All the best with your preparation and exams.
- ▶ All the best for your future.